

ECO2010H1F
Mathematics and Statistics for PhD Students
University of Toronto

Department of Economics
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Term Test 1 - Solutions

Date: August 22, 2025

Time: 10:10 am - 12:00 pm

Question 1

The contrapositive is: If n is odd, then n^2 is odd. Assume n is odd. Then, by definition, there exists an integer k such that

$$n = 2k + 1.$$

Squaring both sides yields

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

This is of the form $2m + 1$ for some integer m , which means n^2 is odd by definition. Since we have proven the contrapositive, the original theorem is true.

Question 2

We will prove that $\{d(x_m, y_m)\}_{m \in \mathbb{N}} \subseteq \mathbb{R}$ is a Cauchy sequence, from which it follows that it converges, because $\mathbb{R}$ is complete. For every $\varepsilon > 0$, since (x^m) and (y^m) are Cauchy sequences, there exists $N > 0$ such that $d(x_n, x_m) < \varepsilon/2$ and $d(y_n, y_m) < \varepsilon/2$ for all $n, m \geq N$. Hence, for $n, m \geq N$ by triangle inequality we have that

$$\begin{aligned} |d(x_n, y_n) - d(x_m, y_m)| &= |d(x_n, y_n) - d(x_m, y_n) + d(x_m, y_n) - d(x_m, y_m)| \\ &\leq |d(x_n, y_n) - d(x_m, y_n)| + |d(x_m, y_n) - d(x_m, y_m)| \\ &\leq d(x_n, x_m) + d(y_n, y_m) \\ &< \varepsilon \end{aligned}$$

which proves that $\{d(x_m, y_m)\}_{m \in \mathbb{N}}$ is Cauchy.

Question 3

Suppose that $c_1, \dots, c_r \in \mathbb{R}$ and

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n = \mathbf{0}$$

Then for every $i = 1, \dots, r$ it holds that

$$\begin{aligned} 0 &= \langle \mathbf{0}, \mathbf{v}_i \rangle \\ &= \langle c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n, \mathbf{v}_i \rangle \\ &= c_i \langle \mathbf{v}_i, \mathbf{v}_i \rangle \end{aligned}$$

since $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ if $j \neq i$. As $\mathbf{v}_i \neq \mathbf{0}$ and hence $\langle \mathbf{v}_i, \mathbf{v}_i \rangle \neq 0$ we must have $c_i = 0$ for every $i = 1, \dots, r$. It follows that $c_1 = \dots = c_r = 0$.

Question 4

Neither, see lecture notes.

Question 5

(a) Given A_1 the period-1 problem is

$$\max_{C_1} C_1^\alpha \quad \text{s.t.} \quad C_1 + A_2 = w_1 + (1+r)A_1.$$

With no bequest motive the household sets $A_2^* = 0$ and consumes all available resources, hence

$$\begin{aligned} C_1^*(A_1) &= w_1 + (1+r)A_1 \\ v_1(A_1) &= \left(w_1 + (1+r)A_1 \right)^\alpha. \end{aligned}$$

(b) Using $A_0 = 0$ and the period-0 budget, $A_1 = w_0 - C_0$. Substituting the period-1 optimal decision and value function into the period-0 objective gives

$$\max_{C_0 > 0} C_0^\alpha + \beta \left[w_1 + (1+r)(w_0 - C_0) \right]^\alpha.$$

Let $S = w_1 + (1+r)w_0$ and

$$D(C_0) = w_1 + (1+r)(w_0 - C_0) = S - (1+r)C_0.$$

The first-order condition (F.O.C.) w.r.t. C_0 is

$$\alpha C_0^{\alpha-1} - \beta \alpha (1+r) D(C_0)^{\alpha-1} = 0.$$

Canceling the positive factor α yields

$$C_0^{\alpha-1} = \beta (1+r) D(C_0)^{\alpha-1}.$$

Let

$$k = \left(\beta (1+r) \right)^{\frac{1}{\alpha-1}} = \left(\beta (1+r) \right)^{-\frac{1}{1-\alpha}},$$

so that the F.O.C. can be written as $C_0 = k D(C_0)$. Substituting $D(C_0) = S - (1+r)C_0$ gives

$$\begin{aligned} C_0 &= k [S - (1+r)C_0] \\ C_0 (1 + k(1+r)) &= kS. \end{aligned}$$

Therefore the closed-form solutions are

$$C_0^* = \frac{kS}{1 + k(1+r)} = \frac{\left(\beta (1+r) \right)^{\frac{1}{\alpha-1}} [w_1 + (1+r)w_0]}{1 + \left(\beta (1+r) \right)^{\frac{1}{\alpha-1}} (1+r)}$$

and, using $C_1^* = D(C_0^*)$,

$$C_1^* = \frac{S}{1 + k(1+r)} = \frac{w_1 + (1+r)w_0}{1 + \left(\beta (1+r) \right)^{\frac{1}{\alpha-1}} (1+r)}.$$

(c) The optimal first-period asset holding is

$$A_1^* = w_0 - C_0^* = w_0 - \frac{kS}{1 + k(1 + r)}.$$

The intertemporal budget is:

$$C_0^* + \frac{C_1^*}{1 + r} = \frac{kS}{1 + k(1 + r)} + \frac{S}{(1 + r)(1 + k(1 + r))} = \frac{S}{1 + r} = w_0 + \frac{w_1}{1 + r},$$

so the household exactly exhausts its lifetime resources.

Question 6

(a) We are given the initial value problem

$$\frac{dx}{dt} = 2t, \quad x(0) = 1.$$

Integrating both sides with respect to t gives

$$x(t) = \int 2t \, dt = t^2 + C,$$

where C is a constant of integration. Applying the initial condition $x(0) = 1$:

$$1 = 0^2 + C \Rightarrow C = 1.$$

Thus, the solution is

$$x(t) = t^2 + 1.$$

(b) We are given the initial value problem

$$\frac{dx}{dt} = kx, \quad x(0) = 5, \quad k > 0.$$

Separating variables:

$$\frac{1}{x} dx = k \, dt.$$

Integrating both sides,

$$\int \frac{1}{x} dx = k \int dt$$

yields

$$\ln |x| = kt + C,$$

where C is a constant of integration. Exponentiating:

$$x(t) = Ae^{kt},$$

where $A = e^C$. Applying the initial condition $x(0) = 5$:

$$5 = Ae^{k \cdot 0} \Rightarrow A = 5.$$

Thus, the solution is

$$x(t) = 5e^{kt}.$$